

Supplementary Materials

Foundation models, deep learning, and statistical methods for disease burden forecasting: a 10-model comparison using GBD 2023 data

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Table of Contents

- Supplementary Table S1. Full forecast matrix: DALY rates at milestone years (ARIMA)
- Supplementary Table S2. Expanding-window cross-validation results
- Supplementary Table S3. Diebold-Mariano pairwise test results
- Supplementary Table S4. Bootstrap 95% confidence intervals for MAE differences
- Supplementary Table S5. Bias analysis by model
- Supplementary Table S6. Augmented Dickey-Fuller stationarity tests
- Supplementary Table S7. Ljung-Box residual autocorrelation tests
- Supplementary Table S8. Residual normality tests
- Supplementary Table S9. Granger causality tests for covariates
- Supplementary Table S10. Leave-one-country-out validation (XGBoost)
- Supplementary Table S11. COVID-19 disruption quantification
- Supplementary Table S12. Model agreement at 2040 forecast horizon
- Supplementary Table S13. Sex-stratified projections for Australia
- Supplementary Table S14. Age-stratified projections for Australia
- Supplementary Table S15. Deaths (mortality rate) projections for Australia
- Supplementary Table S16. Sensitivity analysis: performance by training cutoff
- Supplementary Table S17. Multiple testing correction (Benjamini-Hochberg)
- Supplementary Table S18. Prediction interval width at 2040 (Prophet)
- Supplementary Table S19. Continuous Ranked Probability Score (CRPS)
- Supplementary Table S20. TRIPOD+AI reporting checklist
- Supplementary Methods. TFT architecture and training details
- Supplementary Figure S1. ACF/PACF diagnostic plots for representative series
- Supplementary Figure S2. Covariate correlation heatmap
- Supplementary Figure S3. Model agreement heatmap at 2040
- Supplementary Figure S4. COVID-19 disruption heatmap
- Supplementary Figure S5. Prediction interval width heatmap at 2040
- Supplementary Figure S6. Volatility vs forecast accuracy
- Supplementary Figure S7. Sensitivity analysis: MAE by training cutoff
- Supplementary Figure S8. Age-stratified projection heatmap (Australia)
- Supplementary Figure S9. Sex-stratified projection chart (Australia)
- Supplementary Figure S10. Cross-country divergence plots (depression, DM type 2, dementia)
- Supplementary Figure S11. All-model forecast comparison (Australia, 6 causes)

- Supplementary Figure S12. TFT temporal attention weights

Supplementary Table S1. Full forecast matrix: projected DALY rates per 100,000 at milestone years (ARIMA, retrained on 1990-2023)

Location	Cause	2023 (observed)	2025	% change	2030	% change	2035	% change	2040	% change
AUS	Anxiety disorders	1314.1	1476.0	+12.3	1910.2	+45.4	2344.4	+78.4	2778.7	+111.5
AUS	Asthma	519.3	535.2	+3.1	541.2	+4.2	541.3	+4.2	541.3	+4.2
AUS	Cancer (all)	4345.3	4292.4	-1.2	4330.5	-0.3	4333.2	-0.3	4333.4	-0.3
AUS	Breast cancer	341.0	323.2	-5.2	295.1	-13.4	256.5	-24.8	226.1	-33.7
AUS	Colorectal cancer	478.4	466.2	-2.5	442.8	-7.4	422.7	-11.6	404.3	-15.5
AUS	Liver cancer	198.0	201.5	+1.8	210.4	+6.3	217.3	+9.8	225.0	+13.6
AUS	Lung cancer	733.1	730.2	-0.4	717.0	-2.2	717.1	-2.2	716.7	-2.2
AUS	Prostate cancer	311.1	312.5	+0.4	314.7	+1.1	316.0	+1.6	316.6	+1.8
AUS	CKD	379.3	383.0	+1.0	387.8	+2.2	387.8	+2.2	387.8	+2.2
AUS	COPD	732.0	729.7	-0.3	728.7	-0.5	728.7	-0.5	728.7	-0.5
AUS	Dementia	971.7	999.9	+2.9	1064.7	+9.6	1137.2	+17.0	1215.1	+25.0
AUS	Depression	751.6	752.9	+0.2	755.7	+0.5	755.7	+0.5	755.7	+0.5
AUS	DM type 1	45.2	45.2	-0.1	45.1	-0.4	45.1	-0.4	45.1	-0.4
AUS	DM type 2	612.5	610.7	-0.3	602.6	-1.6	602.6	-1.6	602.6	-1.6
AUS	Hypertension	722.6	71.2	-1.9	69.7	-3.9	69.7	-3.9	69.7	-3.9
AUS	IHD	1375.6	1296.2	-5.8	1113.4	-19.1	948.7	-31.0	798.8	-41.9

Location	Cause	2023 (observed)	2025	% change	2030	% change	2035	% change	2040	% change
AUS	Self-harm	544.6	527.5	-3.1	511.1	-6.2	511.1	-6.2	511.1	-6.2
AUS	Stroke	719.9	681.3	-5.4	604.3	-16.1	547.8	-23.9	497.5	-30.9

Note: Table shows Australia only. Full data for all 5 countries (90 series) is available in the machine-readable file `supplementary_forecast_matrix.csv`. Caution: some individual ARIMA projections at the 2040 horizon are implausible for volatile series (see Supplementary Methods note on long-horizon ARIMA instability).

Supplementary Table S2. Expanding-window cross-validation (ARIMA, 3-year forecast horizon)

Training cutoff	Mean MAE	Median MAE	SD of MAE	Series evaluated
2015	16.56	11.26	21.02	90
2016	16.36	10.64	16.76	90
2017	21.62	14.01	23.92	90
2018	30.60	14.31	39.40	90

Note: Increasing error at the 2018 cutoff reflects inclusion of COVID-19-disrupted years (2019-2021) in the validation window.

Supplementary Table S3. Diebold-Mariano pairwise test results

Model 1	Model 2	DM statistic	p-value	Significant (p<0.05)	Significant (BH-corrected)
ARIMA	Prophet	-2.503	0.012	Yes	Yes
Prophet	TFT	2.469	0.014	Yes	Yes
ARIMA	XGBoost	-1.932	0.053	No	No
XGBoost	TFT	1.955	0.051	No	No
XGBoost	Prophet	0.434	0.665	No	No
ARIMA	TFT	0.065	0.948	No	No

Note: Negative DM statistic indicates Model 1 is more accurate. Benjamini-Hochberg correction applied across all 11 hypothesis tests (6 DM + 5 bias tests).

Supplementary Table S4. Bootstrap 95% confidence intervals for pairwise MAE differences (10,000 resamples)

Comparison	Mean MAE difference	95% CI lower	95% CI upper	Significant	Better model
Ensemble vs ARIMA	-5.07	-7.66	-2.50	Yes	Ensemble
ARIMA vs TFT	-7.56	-12.39	-2.52	Yes	ARIMA
ARIMA vs XGBoost	-10.54	-19.14	-3.73	Yes	ARIMA
ARIMA vs Prophet	-17.84	-25.10	-11.16	Yes	ARIMA
TFT vs Prophet	-10.28	-18.44	-2.85	Yes	TFT

Note: Negative difference indicates the first-named model has lower MAE.

Supplementary Table S5. Bias analysis (one-sample t-test: H0: mean error = 0)

Model	Mean bias (DALYs/100k)	Direction	t-statistic	p-value	Significant
ARIMA	20.95	Under-predicts	5.509	<0.001	Yes
XGBoost	37.74	Under-predicts	7.039	<0.001	Yes
Prophet	7.53	Under-predicts	1.427	0.154	No
TFT	14.12	Under-predicts	3.667	<0.001	Yes
Ensemble	19.95	Under-predicts	5.777	<0.001	Yes

Note: Positive bias indicates actual values exceed predictions (under-prediction).

Supplementary Table S6. Augmented Dickey-Fuller stationarity tests

Of 90 series: - **Stationary at level ($p < 0.05$):** 8 (8.9%) - **Stationary after first differencing:** 59 (65.6%) - **Requiring second differencing or exhibiting structural breaks:** 23 (25.6%)

Full per-series results available in `adf_stationarity_tests.csv`.

Supplementary Table S7. Ljung-Box residual autocorrelation tests (ARIMA models)

Of 90 fitted ARIMA models: - **No significant residual autocorrelation ($p > 0.05$):** 40 (44.4%) - **Significant residual autocorrelation detected:** 50 (55.6%)

Note: Tests conducted at up to 10 lags. The minimum p-value across lags was used to classify each series. Full results in `ljung_box_tests.csv`.

Supplementary Table S8. Residual normality tests

Model	Shapiro-Wilk p	Jarque-Bera p	Normally distributed
ARIMA	<0.001	<0.001	No
XGBoost	<0.001	<0.001	No
Prophet	<0.001	<0.001	No
TFT	<0.001	<0.001	No
Ensemble	<0.001	<0.001	No

Note: Residuals were non-normal for all models, reflecting heavy-tailed error distributions driven by outlier series (particularly mental health conditions).

Supplementary Table S9. Granger causality tests

Covariate	Series tested	Significant ($p < 0.05$)	% significant	Median p-value
Urbanisation rate	90	53	58.9	0.033
Population	90	53	58.9	0.029
Life expectancy	90	44	48.9	0.055
Health expenditure per capita	90	37	41.1	0.091
Physicians per 1,000	90	36	40.0	0.081
GDP per capita (PPP)	90	32	35.6	0.088

Note: Bivariate Granger causality tests (F-test, maximum 3 lags) assessed whether each covariate had predictive value for DALY rates beyond the series' own lagged values.

Supplementary Table S10. Leave-one-country-out validation (XGBoost)

Held-out country	Test observations	MAE	MAPE (%)
Australia	90	39.48	2.65
New Zealand	90	25.85	3.50
Canada	90	17.11	1.75
United Kingdom	90	54.38	3.50
United States	90	37.46	2.39
Mean	—	34.86	2.76

Note: Standard XGBoost (all countries in training) achieved MAE 53.35. The lower LOCO MAE (34.86) indicates that cross-country spatial prediction is easier than temporal forecasting; country-specific temporal dynamics, not cross-country patterns, are the primary source of forecast error.

Supplementary Table S11. COVID-19 disruption classification

Of 90 series: - **Permanently shifted (>5% deviation from pre-2020 trend in 2023): 45 (50.0%)** - **Recovered (<5% deviation): 45 (50.0%)**

Five most positively disrupted series (increased burden):

Series	2020 deviation (%)	2023 deviation (%)	Status
GBR depression	+23.6	+22.7	Shifted
USA anxiety	+23.1	+49.1	Shifted
USA depression	+21.2	+22.4	Shifted
GBR anxiety	+20.0	+37.8	Shifted
CAN depression	+18.0	+17.3	Shifted

Five most negatively disrupted series (decreased burden):

Series	2020 deviation (%)	2023 deviation (%)	Status
GBR COPD	-7.1	-3.7	Recovered
AUS COPD	-7.0	+1.8	Recovered
CAN COPD	-6.7	-2.0	Recovered
NZL self-harm	-6.7	-5.0	Recovered
NZL stroke	-5.5	+8.4	Shifted

Full results in covid_disruption.csv.

Supplementary Table S12. Model agreement at 2040 forecast horizon

Inter-model coefficient of variation (CV) across ARIMA, XGBoost, and Prophet at 2040:

Confidence level	CV range	Number of series	Percentage
High	<10%	14	15.6%
Moderate	10-25%	32	35.6%
Low	25-50%	27	30.0%
Very low	>50%	17	18.9%

Full per-series results in model_agreement_2040.csv.

Supplementary Table S13. Sex-stratified projections for Australia (ARIMA, 2023-2040)

Cause	Male % change	Female % change	Sex gap (pp)
CKD	+2.2	-48.9	51.1
Breast cancer	+2.3	-33.5	35.8
Dementia	+34.8	+16.0	18.8
Stroke	-15.5	-33.7	18.2
IHD	-39.4	-54.2	14.9
Self-harm	+1.7	-5.3	7.0
Hypertensive HD	-0.8	-4.0	3.3
Colorectal cancer	-13.2	-14.6	1.4
COPD	-0.9	-1.7	0.8
DM type 1	+0.3	-0.4	0.7
Cancer (all)	+0.3	+0.2	0.1
Depression	-0.7	+0.6	-1.2
DM type 2	-5.1	+1.1	-6.2
Asthma	-7.2	+2.8	-10.0
Lung cancer	-30.9	+2.6	-33.5
Liver cancer	-6.4	+43.2	-49.6
Anxiety	+115.7	+357.6	-241.9

Note: Sex gap = male % change minus female % change. Positive gap indicates males worsening relative to females. Prostate cancer excluded (sex-specific condition).

Supplementary Table S14. Age-stratified projections for Australia (ARIMA, 2023-2040, % change)

Cause	0-14 years	15-49 years	50-69 years	70+ years
Anxiety	+3570.4	+489.9	+89.1	+184.2
Asthma	+75.9	-6.1	-13.8	-27.7
Cancer (all)	-0.7	-56.0	-43.3	-3.5
Breast cancer	—	-75.1	-37.9	-9.7
CKD	+38.5	+0.1	-4.3	+0.8
Dementia	—	-45.1	-4.6	+3.9
Depression	+1.0	-0.1	+0.2	-0.5
DM type 1	+12.0	-0.3	+0.1	-5.0
DM type 2	—	-8.1	-2.5	-2.3
IHD	—	-23.4	+34.5	-59.1
Self-harm	+2.4	-7.4	-2.3	-0.1
Stroke	-39.2	-41.2	-42.2	-41.6

Note: “—” indicates the cause is rare or absent in that age group (base rate near zero). Only causes with meaningful burden in the age group are shown. Full results in *age_stratified_changes.csv*.

Supplementary Table S15. Mortality rate projections for Australia (ARIMA, deaths per 100,000)

Cause	Deaths/100k (2023)	Deaths/100k (2040)	Change (%)
Liver cancer	9.2	13.4	+45.3
Dementia	54.7	66.6	+21.7
Cancer (all)	213.6	234.8	+9.9
Prostate cancer	18.1	19.5	+7.9
Asthma	1.7	1.8	+4.3
CKD	20.6	21.4	+3.7
Lung cancer	37.2	37.6	+1.0
Colorectal cancer	24.2	24.4	+0.6
DM type 2	19.3	19.4	+0.4
Breast cancer	14.0	14.0	-0.0
COPD	35.6	34.9	-1.9
DM type 1	0.4	0.4	-2.1
Hypertensive HD	4.8	4.6	-2.8
Self-harm	12.3	11.9	-3.4
IHD	83.5	58.7	-29.7
Stroke	40.8	22.2	-45.7

Note: ARIMA fitted to GBD 2023 death rates (per 100,000). Anxiety and depression are excluded as they have negligible

direct mortality in GBD. Directional consistency with DALY projections was observed for all causes.

Supplementary Table S16. Sensitivity analysis: mean MAE by training cutoff

Training cutoff	Forecast horizon (years)	ARIMA MAE	XGBoost MAE
2016	7	37.87	48.22
2017	6	36.68	53.84
2018	5	42.83	53.35
2019	4	49.21	61.72
2020	3	169.85	62.90

Note: ARIMA performance degraded sharply when trained through 2020, as COVID-19 disruptions in the training data distorted autoregressive trend estimates. XGBoost was more robust due to its non-parametric feature-based architecture.

Supplementary Table S17. Multiple testing correction (Benjamini-Hochberg FDR)

Test	Raw p-value	BH threshold	Significant (raw)	Significant (BH-corrected)
Bias: ARIMA	<0.001	0.005	Yes	Yes
Bias: XG-Boost	<0.001	0.009	Yes	Yes
Bias: Ensemble	<0.001	0.014	Yes	Yes
Bias: TFT	<0.001	0.018	Yes	Yes
DM: ARIMA vs Prophet	0.012	0.023	Yes	Yes
DM: Prophet vs TFT	0.014	0.027	Yes	Yes

Test	Raw p-value	BH threshold	Significant (raw)	Significant (BH-corrected)
DM: XG- Boost vs TFT	0.051	0.032	No	No
DM: ARIMA vs XG- Boost	0.053	0.036	No	No
Bias: Prophet	0.154	0.041	No	No
DM: XG- Boost vs Prophet	0.665	0.045	No	No
DM: ARIMA vs TFT	0.948	0.050	No	No

Note: All 6 tests significant at the uncorrected level remained significant after BH-FDR correction at $q < 0.05$.

Supplementary Table S18. Prediction interval width at 2040 (Prophet)

Summary statistics for Prophet 80% prediction interval width at the 2040 horizon:

Statistic	PI width (DALYs/100k)	PI width (% of forecast)
Mean	271	55%
Median	90	17%
Minimum	1	2%
Maximum	2,755	1,165%

Note: Extreme widths (>100% of forecast) were concentrated in GBR series with volatile post-COVID trajectories. Full per-series results in `pi_width_2040.csv`.

Supplementary Table S19. Continuous Ranked Probability Score (CRPS)

Model	CRPS (DALYs/100k)
TFT (5-quantile)	45.90
Prophet (3-quantile approximation)	51.73

Note: Lower CRPS indicates better probabilistic forecast calibration. TFT outperformed Prophet despite both having poor interval coverage, indicating that TFT's quantile predictions were closer to the true distribution even though neither achieved nominal coverage rates.

Supplementary Figures

Supplementary Figure S1. ACF and PACF diagnostic plots for six representative series

File: outputs/figures/acf_pacf_diagnostics.png

Autocorrelation function (ACF) and partial autocorrelation function (PACF) plots for six representative series spanning different disease categories and temporal dynamics. Dashed lines indicate 95% significance bounds. Strong positive autocorrelation at multiple lags (typical of trending series) is evident in most series, confirming the need for differencing. The PACF structure (significant at lag 1, decaying thereafter) supports ARIMA model specifications of order $p=1-3$.

Supplementary Figure S2. Spearman correlation between covariates and DALY rates by cause

File: outputs/figures/covariate_correlation_heatmap.png

Correlation heatmap showing Spearman rank correlations between each World Bank covariate and DALY rates, computed separately for each of the 18 disease causes. Positive correlations (red) indicate covariates associated with higher burden; negative correlations (blue) indicate covariates associated with lower burden. Notable patterns include strong negative correlation between GDP/health expenditure and cardiovascular diseases (reflecting the epidemiological transition in wealthier economies) and positive correlation between these economic indicators and mental health conditions.

Supplementary Figure S3. Model agreement heatmap at 2040

File: outputs/figures/model_agreement_heatmap.png

Coefficient of variation (CV%) across ARIMA, XGBoost, and Prophet forecast values at 2040, by cause and country. Green cells indicate high model agreement ($<10\%$ CV); red cells indicate high disagreement ($>50\%$ CV). GBR series show the highest disagreement, driven by volatile post-COVID trajectories that different model architectures extrapolate differently.

Supplementary Figure S4. COVID-19 disruption heatmap

File: outputs/figures/covid_disruption_heatmap.png

Percentage deviation of observed 2020 DALY rates from the 2015-2019 linear trend extrapolation, by cause and country. Red cells indicate burden exceeded expectations (positive disruption); blue cells indicate burden fell below expectations

(negative disruption). Depression and anxiety show consistent positive disruption across all countries. COPD shows consistent negative disruption, likely reflecting reduced respiratory infections during pandemic restrictions.

Supplementary Figure S5. Prediction interval width at 2040

File: outputs/figures/pi_width_2040.png

Prophet 80% prediction interval width at 2040, expressed as a percentage of the point forecast, by cause and country. Values exceeding 100% indicate prediction intervals wider than the forecast itself, rendering them uninformative for planning purposes.

Supplementary Figure S6. Historical volatility vs forecast accuracy

File: outputs/figures/volatility_vs_accuracy.png

Scatter plot of historical volatility (standard deviation of year-over-year percentage changes, 1990-2023) versus ARIMA forecast MAE on the test set (2019-2023), with points coloured by country. Spearman $r=0.196$ ($p=0.064$). The weak positive association suggests that historical volatility is a limited predictor of forecast accuracy; structural breaks (particularly COVID-19) contribute substantial error independently of historical variability.

Supplementary Figure S7. Sensitivity analysis: MAE by training cutoff

File: outputs/figures/sensitivity_cutoff.png

Mean MAE for ARIMA and XGBoost models across five training cutoff years (2016-2020), all forecasting through 2023. ARIMA performance degrades sharply when trained through 2020 (MAE 169.85), while XGBoost remains relatively stable (MAE 62.90), demonstrating the vulnerability of autoregressive models to training-period disruptions.

Supplementary Figure S8. Age-stratified projection heatmap (Australia)

File: outputs/figures/age_stratified_heatmap.png

Projected percentage change in DALY rates from 2023 to 2040 for Australia, by cause and age group (0-14, 15-49, 50-69, 70+). The extreme projected increase in anxiety for the 0-14 age group (+3,570%) is driven by a steep recent trend from a low base (3.6 DALYs/100,000 in 2023) and should be interpreted as indicative of trajectory direction rather than a precise point estimate.

Supplementary Figure S9. Sex-stratified projections (Australia)

File: outputs/figures/sex_stratified_projections.png

Projected percentage change in DALY rates from 2023 to 2040 for Australia, by cause and sex. Blue bars represent males; red bars represent females. The largest sex disparities are observed for liver cancer (female +43.2%, male -6.4%), anxiety (female +357.6%, male +115.7%), and CKD (male +2.2%, female -48.9%).

Supplementary Figure S10. Cross-country divergence plots

Files: outputs/figures/divergence_depression.png, divergence_dm_type2.png, divergence_dementia.png

Historical (1990-2023, solid lines) and projected (2024-2040, dashed lines) DALY rate trajectories for depression, diabetes mellitus type 2, and dementia across all five countries. These plots illustrate where country-specific trajectories diverge, highlighting conditions where Australia’s projected path differs most from peer country means.

Supplementary Figure S11. All-model forecasts for Australia (6 key causes)

File: outputs/figures/aus_all_models_forecast.png

Historical data (black solid line) and 2024-2040 forecasts from ARIMA (blue), XGBoost (green), and Prophet (orange) for six key causes in Australia. Model disagreement is most visible for depression (divergent forecast directions) and dementia (divergent growth rates), while IHD shows strong model consensus on continued decline.

Supplementary Figure S12. TFT temporal attention weights

File: outputs/figures/tft_temporal_attention.png

Mean temporal attention weight across all prediction windows in the validation set. The TFT’s self-attention mechanism assigns weights to each position in the encoder (historical input), indicating which past time points the model considers most informative for generating forecasts.

Supplementary Table S20. TRIPOD+AI Reporting Checklist

Item	Section	Page
Title identifies as prediction model study	Title	1
Structured abstract	Abstract	1
Background and rationale	Introduction	2
Objectives	Introduction (final paragraph)	2
Source of data	Methods: Study design	2
Participants/setting	Methods: Study design (5 countries, 18 causes)	2
Outcome defined	Methods: Study design (DALY rates)	2
Predictors/covariates	Methods: Study design (6 World Bank covariates)	2
Sample size	Methods: Data preparation (90 series, 34 obs each)	3
Missing data handling	Methods: Study design (linear interpolation)	3
Model development	Methods: Forecasting models	3
Model performance measures	Methods: Statistical validation	4
Statistical methods	Methods: Statistical validation	4
Risk groups	Methods: Subgroup analyses (sex, age)	4
Model performance results	Results: Model comparison	5

Item	Section	Page
Model updating	Methods: Production forecasts (retrained on full data)	4
AI: Model architecture	Methods: TFT paragraph	3
AI: Training procedure	Methods: TFT paragraph (loss, lr, early stopping)	3
AI: Hyperparameters	Methods: TFT paragraph (hidden size, heads, dropout)	3
AI: Computing environment	Methods: Software (Python 3.12, CPU, PyTorch 2.2)	4
AI: Code availability	Methods: Software (repository URL)	4
Limitations	Discussion: Limitations	7
Interpretation	Discussion	6-7
Implications	Discussion: final paragraphs	7
Reporting checklist	This table (Supplementary Table S20)	Supp

Supplementary Methods. TFT Architecture and Training Details

Model architecture

The temporal fusion transformer comprised the following components: - **Input embeddings:** Entity embeddings for static categorical variables (location: 5 categories; cause: 18 categories), each embedded to dimension 64. - **Variable selection networks:** Separate networks for static, encoder (time-varying unknown), and decoder (time-varying known) inputs. Each used a GRN (gated residual network) with ELU activation to produce softmax-normalised importance weights. - **LSTM encoder-decoder:** Single-layer LSTM with hidden size 64, processing the encoder sequence (up to 20 time steps) and decoder sequence (up to 5 time steps). - **Multi-head attention:** 4 attention heads operating over the LSTM encoder outputs, with interpretable attention weights providing temporal importance scores. - **Output layer:** Position-wise feed-forward network producing 5 quantile predictions (0.10, 0.25, 0.50, 0.75, 0.90) per forecast horizon step.

Target normalisation

Each time series was independently normalised using a softplus group normaliser, which computes per-group (per series) location and scale parameters. This ensures the model sees comparable magnitudes across series with vastly different scales (e.g., IHD at ~1,376 vs DM type 1 at ~45 DALYs/100k).

Training procedure

- **Loss function:** Quantile loss across 5 quantiles, summed over all forecast horizon steps
- **Optimiser:** Adam with initial learning rate 1e-3, reduced on plateau (patience 5 epochs, factor 0.1)
- **Gradient clipping:** Maximum gradient norm 0.5

- **Early stopping:** Patience 10 epochs monitoring validation loss
- **Batch size:** 64
- **Training device:** CPU (Intel, no GPU acceleration)
- **Training duration:** Approximately 35 epochs (early stopping triggered), ~10 minutes on CPU
- **Total parameters:** 306,255

Convergence

Training converged after approximately 35 epochs with validation loss stabilising. No evidence of overfitting was observed (training and validation loss curves tracked closely after epoch 15).

ARIMA order selection

The most common ARIMA orders selected across the 90 series were: (0,2,3) for 19 series, (1,1,3) for 17 series, (3,1,3) for 10 series, and (0,1,3) for 10 series. The predominance of $d=1$ and $d=2$ confirms the non-stationarity documented by ADF testing. The high q values ($q=3$ in most models) suggest significant moving-average structure in the differenced series.

Note on long-horizon ARIMA instability

Some ARIMA models produced implausible long-horizon (2040) forecasts, including negative values (GBR asthma) and extreme exponential growth (CAN anxiety +7,544%, GBR depression +323%, GBR IHD +157%). These arise from ARIMA's tendency to extrapolate recent volatile trends (particularly post-COVID disruptions) indefinitely. For policy interpretation, the ensemble forecasts and the model agreement analysis (Supplementary Table S12) should be preferred over individual ARIMA projections for series with high model disagreement (CV >50%).